

THE  
SEVENTH  
WAVE

*or, an*  
ESCAPE  
*from*  
PERFORMED FUTURE

*Being an Essay upon the Theories of Futures  
Studies, and whether they shape the Futures  
they describe.*

By VELI VIRMAJOKI

MMXXVI

# The Seventh Wave

## Escape from Performed Future

Veli Virmajoki · draft essay, September 2026

**Abstract.** This essay asks two questions and argues that they are, in the end, one question. A scene from the film *Papillon* (1973) is used to hold them together. First: what are the theories of futures studies theories of? The talk about *theory* in the field is a mess. I argue that the mess becomes smaller when we look at the roles that theories play in a study and leave aside the question what theories are as such – here I use the distinction of Lukka and Vinnari between *domain theory* and *method theory* – and when we name the object that the field’s own *domain theories* would be about. That object is the *possibility space*. Second: do the theories, models and pictures of the field shape the futures they describe? I lean on the performativity literature, especially on MacKenzie and Millo’s study of option pricing, and argue three things: that performativity is a family of mechanisms and no single thing, that futures studies has a variety of its own (I call it *modal performativity*), and that every claim of performativity is a causal claim which needs counterfactual evidence. The two questions meet because the *possibility space* contains accounts of itself. The lens is in the picture, so to say.

### 1. Devil’s Island

In *Papillon* (1973), Franklin J. Schaffner’s film, Steve McQueen plays a convict who tries for years to escape from the French penal colony in Guiana. In the end of the film he is on Devil’s Island. It is a rock with cliffs and currents around it, and everyone agrees that nobody escapes from there. We need to notice that the agreement matters as much as the rocks do. The island has an image of its own future (no one leaves), and this image tells to the prisoners which futures are not worth of attempting. Most of them act accordingly.

Papillon does something else. He sits on a cliff and watches the sea, one day after another, until he notices a regularity that nobody else has cared to see. The waves come in a rhythm, and every seventh wave is strong enough to carry something out to the open water. However, he does not trust the observation immediately. He tests it: he throws sacks of coconuts into the surf and watches where they go. Then he jumps. His friend Dega (played by Dustin Hoffman) has watched the same waves from the same cliff. Dega stays.

I keep returning to that cliff. Why? Because the scene contains, in a few minutes, most of the problems this essay discusses about. First, there is a space of possible futures, and the space is no mere list. It has a structure: some futures are closed, some are only conceivable, and some become plausible once a route is found. Second, there is a present that has to be reconstructed before anything can be projected from it – the same sea looked like a wall to everyone except one man. Third, there is a regularity that does not care whether anyone knows it. The waves would have come in sevens even if Papillon had never counted

them. Finally, there is an image of the future, nobody escapes, and this image works for the very reason that people believe it. This last kind of thing is what I call performative in the later sections. The sea did not read the forecast of Papillon. Prisoners and guards read theirs, and so do markets, organisations, societies, and so on.

It follows two questions which, I will argue, turn out to be one. The first question is: what are the theories of futures studies theories of? The talk about “theory” in the field is a mess, and the mess is not merely about terminology. The second question is: do the theories, models and pictures of futures studies shape the futures they describe? And if they do, what follows from this to how we should evaluate them? The two questions meet because the object of futures studies, as I understand it, contains accounts of itself. The lens is in the picture. What that even means is not clear at this point, of course, and it takes the rest of the essay to make sense of it.

A word about where I stand, since it affects everything below. I come to futures studies from the philosophy of science, and in a sense I remain a guest in the field (although a guest who has spent rather long time reading the house rules). The philosophy of science I practise is modest and oriented to practice. It does not begin from what the future “really is”. It begins from what researchers do when they study the future, and it attempts to make that doing clearer (Virmajoki 2026a). Readers who want the metaphysics of time will be disappointed, no doubt about that. Readers who want to know what their scenarios are for will perhaps be less disappointed, or so I hope.

## **2. Too little theory, and too much**

Two complaints about futures studies are made often. They are made seriously, and sometimes the same people make both. The first complaint says that the field has too little theory. According to this complaint, futures studies is a set of methods and practices, and it has never developed the kind of tested and cumulative theory that mature sciences have. For example, Fergnani and Chermack (2021) went as far as to diagnose a resistance to scientific theory in futures and foresight. The second complaint says the opposite: the field has too much theory. There are grand frameworks – *integral theory*, *poststructuralism*, this or that philosophy of consciousness and history – and they compete over which of them is the most comprehensive and which one can “subsume” the others. Anyone who has read the exchanges between *Integral Futures* and *Causal Layered Analysis* in *Futures* (vol. 40, no. 2, 2008; vol. 42, no. 2, 2010) knows the tone.

Both complaints point at something real. However, they cannot both be true in the same sense of the word. A field cannot have too little and too much of the same thing at the same time. So we have two options. Either one of the

complaints is simply wrong, or the word “theory” does different jobs in them. I think the second option is the correct one. Moreover, I think the ambiguity is costly, as we will see.

Consider what the word covers in the ordinary talk of the field. It covers grand frameworks such as *integral theory*. It covers methods too: people speak of *scenario theory*, or say that *CLA* is their theory. It covers pictures and typologies, for example the *futures cone* and the distinction between possible, probable and preferable futures. It covers knowledge that is borrowed from other fields (*demographic transition*, *learning curves* in energy technology) when such knowledge is used in the building of futures. It covers large empirical-historical theses, such as the claim of Fred Polak that the images of the future which a civilisation holds steer where that civilisation goes (Polak 1973). And it covers reflections on the field itself, such as the attempt of Wendell Bell to lay out the foundations of the field (Bell 2009). That is one word and at least six jobs.

Why should anyone care? Because the ambiguity has consequences to evaluation. Consider a student who tells me that her study “uses *CLA* as its theory”. I cannot tell what would count as success in that study. Does she want to show that *poststructuralism* is fruitful? Or that it is true? Does she want to say something new about the futures of urban mobility, let’s say? Or something about how *CLA* works? Each of these aims comes with different standards. The same holds for the debates between the grand frameworks. When one framework claims to subsume another, we need to ask what kind of claim that is. Is it a claim about logical entailment, about explanatory hierarchy, or about how practitioners position themselves in a community? The debates rarely tell (Virmajoki 2026a). To be blunt, they produce a great deal of heat and very little light. And the complaint that the field “lacks theory” can be neither confirmed nor refuted as long as nobody says which kind of theory the field is supposed to be lacking.

When one faces a mess like this, there is a temptation to legislate: to declare what theory really is and to sweep everything else away. I resist the temptation. The philosophy of science has tried legislating before (with falsifiability, with paradigms, with research programmes), and what it learned is that grand definitions of science break down once they meet the plurality of actual scientific practices. A more productive move, it seems to me, is to ask what a theory does in a given study, i.e., what role it plays there. This is a question one can actually answer, one case at a time. Luckily for us, it is also a question that has already been answered with care in a neighbouring field.

### 3. Roles, not kinds

Kari Lukka and Eija Vinnari (2014) wrote about management accounting, which is a field with its own version of our worry. Their proposal is simple – so simple that it is easy to underestimate. A theory can play two rather different roles in a study. A *domain theory* is a body of knowledge about a substantive topic area of the field. In accounting, this means knowledge about budgeting, cost accounting, performance measurement, and so on. A *method theory* is a conceptual system at meta-level, a lens as it were, which is typically imported from another discipline and used to study the substantive issues of the domain. *Actor-network theory* and *Foucault's theory of power and knowledge* are used as *method theories* in accounting research.

The important point is that the roles are relative. No theory is a *domain theory* or a *method theory* as such. Foucault's *theory of power and knowledge* is a *domain theory* in sociology. However, when accounting researchers borrow it, it is a *method theory*. The theory is the same; the job is different. This may look like a small observation, but it dissolves surprisingly much confusion. Why is that? Because it changes the question we ask. We no longer ask “what kind of thing is this theory?”. We ask “what is this theory doing here?”.

Lukka and Vinnari studied how *actor-network theory* has been used in accounting research, and the study contains three observations that are worth to carry over to futures studies. First, most studies use the lens instrumentally, in order to say something about the domain. Only few try to develop the lens itself, and that is perfectly acceptable. Second, some studies have what Lukka and Vinnari call an unclear theoretical ambition. Such studies apply a lens and tell an empirical story, but one cannot say whether the contribution is meant to land in the domain or in the lens. Lukka and Vinnari borrow a phrase from Vaivio (2008, p. 77) here: such studies show that “it fits, it fits!” – the lens fits the material – and they show nothing else. Third, lenses leave traces, and this is the most interesting observation of the three. Over two decades, concepts from *actor-network theory* such as “action at a distance” and “fabrication” became part of the ordinary vocabulary of accounting research. They were used without reference to where they came from, and they were even treated as received wisdom that should be challenged. So a *method theory* can reshape how a field sees its own domain while the *method theory* itself is not modified at all.

When we apply this to futures studies, much of the complaint about too much theory becomes understandable. In these terms, *integral theory* and *poststructuralism* are *method theories*. *Causal Layered Analysis* says as much in the subtitle of its founding article: *poststructuralism as method* (Inayatullah 1998). To be sure, there is nothing wrong with a lens. The trouble begins when lenses compete with other lenses and nobody says what the competition is

supposed to decide in the field's own domain. From this angle, much of the exchange between *Integral Futures* and *CLA* looks like a dispute between *method theories* where no *domain theory* is in sight. It is the unclear theoretical ambition again, but now in the scale of a whole field.

Lukka and Vinnari describe also a precedent that should make futures scholars somewhat uneasy. Around the turn of the millennium, management accounting research was criticised for having produced little theory. The remedy Zimmerman proposed was to adopt economics as the underlying discipline for theory building (Zimmerman 2001). Lukka and Vinnari warn that when one broad *method theory* becomes the foundation of a field, research tends towards homogeneity: the lens becomes the field. Futures studies has heard a similar diagnosis and a similar remedy (Fergnani & Chermack 2021). Develop and test one kind of theory (a theory of foresight and its effects), and the field will finally become scientific. The content of the cure is different from the accounting case, but the shape of the cure is the same. The cost would be similar, too. A great deal of what futures studies actually does is design-oriented, critical, interpretive or participatory work, and such work would fall outside a field that is unified around one kind of theory.

However, at this point the tool stops helping us, and a deeper problem comes into view. The distinction between *domain theory* and *method theory* presupposes that there is a domain, i.e., a topic area that the field can call its own. Lukka and Vinnari defend the existence of an accounting domain in a pragmatic manner: there are journals that reject manuscripts which do not address accounting. What does this argument show? It shows that a practice exists, a community that polices its borders. It does not show that there is an object. And in futures studies it is exactly the existence of an object that is in question.

#### **4. What is left for futures studies?**

The problem is easy to state and hard to get rid of. Every domain of life that futures studies examines is already studied by some special science. Economists study economic futures, climate scientists study climate futures, demographers study population futures, and educational researchers study the futures of education. If every future has a home discipline already, what is left for a separate field? We need an answer. Without one, futures studies has only *method theories*, that is, lenses and procedures which are applied to the topics of other fields. It would then be what Lukka and Vinnari describe as one extreme view of accounting: a purely empirical field that is studied with imported theories. It would be a toolbox and nothing more (Virmajoki 2026b).

Three familiar replies circulate, and each of them captures something true about how futures work is done. The first reply says that futures studies extracts

forward-looking insight from experts, as the *Delphi method* does. However, structured expert elicitation is done inside the home sciences too (think of climate assessments or technology roadmaps), and a technique can be carried anywhere. The second reply says that futures studies takes a longer view than the special sciences. However, there is no single futures horizon. Practitioners work with widely varying time horizons, depending on the subject and purpose of the inquiry (Brier 2005). Moreover, what makes futures work difficult is not connected to time scales in any simple way: some near futures are deeply uncertain, and some distant ones are tightly constrained. The third reply says that futures studies includes the perceptions and expectations of lay people. Often it does, and admirably. But citizens' panels and deliberative forums exist inside the home sciences as well. So each reply tells us what futures studies does. None of them tells what it studies.

The distinction I find most helpful here is the distinction between an objective and an object. Methods, the improving of methods and the support for decision-making are objectives. They are things that are pursued, and they are worth pursuing, to be sure. An object is that on which the objectives are pursued. The field has never lacked objectives. What it has lacked is an object with a clear name. This is, once again, the unclear theoretical ambition of Lukka and Vinnari, now at the level of a whole field.

My proposal is that a proper object of futures studies is the *possibility space* (Virmajoki 2026b). A *possibility space* is a structured whole. It is not a list of alternatives. It has four features.

First, it has a *modal architecture*. It contains different kinds of possibility at once: predictable continuations, plausible developments, merely possible scenarios, alternatives that arise when we question what we believe and value, desirable futures, and futures that are only conceivable. These are not the same thing, and much of the confused talk about futures comes from treating them as if they were. To be sure, the special sciences deal with modality all the time, in laws, counterfactuals and projections. However, none of them takes as its subject the layered structure where several modalities interact. These modal notions are associated with epistemic claims, and they carry both practical and philosophical weight. That weight can only be assessed, if modal vocabulary is communicated clearly.

Second, it has a *multi-domain composition*. A *possibility space* is assembled from the outputs of many fields, and the assembling is no trivial thing. The regularities of demography, economics and political science have different scopes and different ranges of validity. Whether they can be combined at all is itself a question. So the assembling is part of the inquiry. It is not something one does before the inquiry begins.

Third, it starts from a *contested present*. The present from which futures are projected is not given. What is the current geopolitical situation? Is this or that technology on the verge of a breakthrough, or is it oversold? Is democracy in crisis, or is it adapting? These are disputed matters, and which present one reconstructs changes which futures are open. We need to be careful here. I am not claiming that once the present is settled, forecasting would be licensed; in complex systems even a precise knowledge of initial conditions would not give that. The claim is that the mapping of a *possibility space* is a different act than the extrapolating of a trend.

Fourth, it is reflexive. Accounts of the *possibility space* partly shape the space. Expectations, visions and scenarios feed back to the actions that bring futures about or prevent them.

The four features are not independent from each other. Which futures count as plausible depends on which domain inputs are taken as authoritative. Which inputs are relevant depends on how the present is reconstructed. How the present is reconstructed depends, in part, on what futures we expect, hope for or fear. And the accounts that shape the future are themselves located in the modal layers: a vision is an account that aims at what counts as desirable, while an alternative scenario is an account that questions what counts as plausible. A study that handles only one or two of the features is studying something else. This is why, I think, the home sciences do not study this object. It is also why futures studies keeps being reinvented under new names (possibility studies, opportunity spaces, and so on) whenever someone needs it. One could argue that the growing number of neighbours is a counterexample to the claim that the home sciences leave something unstudied. It seems to me that it is the opposite. It is corroboration.

Let us return to the cliff for a moment. The scene is almost a diagram of a *possibility space*. An escape was conceivable but dismissed as impossible. It becomes plausible once a route can be described (a regularity together with a mechanism), and it becomes credible once the route has been tested with coconuts. No sketch of the path, no plausibility. Nothing in the sea changed. What changed was the reconstruction of the present and the knowledge that was brought to bear on it. The island's image of the future had done part of the closing of the space; a closer attention to the present opened the space again. Moreover, the choice of Dega shows that the modal layers do not collapse to one another. The same future was plausible for both men, but it was desirable for only one of them. The years on the island had shaped what Dega wanted – his garden, his animals, a kind of safety. Philosophers of justice speak of preferences that have adapted to poor conditions, and this is such a case. A *possibility space* without its desirability layer would miss the most human thing in the scene.



The most familiar picture of the *possibility space* is the *futures cone*. There is one present at the apex, and the futures widen outward from it: probable futures are inside plausible ones, plausible ones are inside possible ones, and preferable futures cut across them (Hancock & Bezold 1994; Voros 2003). Pedagogically the cone is excellent, as it shows at one glance that futures thinking is more than prediction. However, the cone can also be taken as a theory – and it is one, a theory of the modal structure of futures – and as a theory it is wrong in at least three ways. First, it suggests that the different kinds of possibility coexist in fixed shares at any future date, as if one could pick 2050 and ask how much of it is plausible. In fact, which kinds of possibility are available depends on how far one is willing to go and what one is satisfied with. Second, the cone draws a settled apex, while the present is contested, as we have seen. Third, it suggests a smooth widening that never turns back, while *possibility spaces* also shrink when commitments harden and options close. Consider a nuclear war. After it, the plausible and the desirable regions would not widen. They would wither to almost nothing.

Once the object has a name, the division of labour between futures studies and the special sciences becomes clearer. The special sciences supply content, models and constraints.

Futures studies studies the space that those constraints together leave open. What we do is to navigate a space. We are not searching for a point in it. In general, the idea that understanding can be defined as the ability to navigate spaces of possibility provides one suitable philosophy for futures studies. According to Soazig Le Bihan (2017), a person has (modal) understanding of a phenomenon when that person possesses the know-how to move around within some region of the space of possibility associated with that phenomenon. We do not have understanding when we can see possibilities here and there, but we have to competently navigate a space of possibilities. Futures studies, on this view, is not an exotic supplement to ordinary inquiry but a natural expression of the drive toward understanding.

## **5. A tidier map – and where theory hides**

Now we can sort the mess, or at least some of it. Theory in futures studies plays at least five roles. First, a theory can be a lens that is borrowed from elsewhere. This is the *method theory* of Lukka and Vinnari. Second, it can be an input: knowledge from a home science, such as *demographic transition* or *learning curves*, whose outputs feed the *possibility space* as constraints. Lukka and Vinnari have no slot for this role, and the reason is that accounting does not routinely assemble the outputs of other sciences. Third, it can be a theory of our object, i.e., a claim about *possibility spaces* and their structure. This is what a *domain theory* of futures studies would be. The cone is one such theory, the

thesis of Polak is another, and my account of the four features is a third. Fourth, a theory can be built into methods: it then consists of assumptions that are embodied in the design of a method and not stated as propositions. And finally, a theory can be about the field. Here belong the accounts of what futures studies is, from the foundations of Bell to the essay you are reading.

With this map, the two complaints stop contradicting each other. “Too much theory” is a complaint about lenses. “Too little theory” is a complaint about theory of our own object, and here the complaint is partly right. However, it is not right for the reason that is usually given. The usual reason is that the field is unscientific. I think the reason is another one: the field lacks explicit *domain theory* because it never named its domain. Its best-known *domain theory* is the cone, and the cone has survived largely because nobody has treated it as a theory that could be false.

The fourth role deserves more attention than it gets. Much of what futures studies knows is never written down as a sentence. It is built into the methods of the field. Consider a scenario team that insists on four scenarios where one forecast was asked for. The insistence embodies a claim: a single forecast is fragile, and the most common failure in reasoning about the future is to get the frame wrong, which is far worse than getting the details wrong. Or consider a Delphi study that reports the spread of expert opinion in addition to the median. The format embodies a claim that disagreement is informative and that consensus does not guarantee correctness. When *backcasting* starts from a desired end state, it embodies the claim that futures can be made and not only anticipated. And when a *morphological analysis* defines its dimensions, it embodies the recognition that the dimensions are themselves assumptions, so that relevant configurations may lie outside the space that is defined in that way.

I have called this embodied knowledge *implicit problem recognition* (Virmajoki, forthcoming). We should notice that it is not reflexivity in general, as every mature field has that. It is expertise in a particular constellation of problems. These problems arise when reasoning is directed at something that does not exist yet and may never exist: the space is open, the object is absent, there is a pull towards one expected trajectory, and the inquiry is entangled with what is inquired into. The choice of a method is therefore always an epistemic choice. It decides what will count as knowledge in the situation at hand.

This explains also why a theory cannot simply be imposed to the field from above. Each approach in futures studies is a package, or an ecosystem as it were, of goal, object, method and theory, and in the package each part supports the others. For example, remove from Delphi the assumption that relevant

knowledge is distributed across experts. The aggregation then loses its point, although the procedure formally remains in place. Or impose *poststructuralism* on the same ecosystem. Now the change is more sweeping: goals, objects and methods must all be described anew, and the statistics become unintelligible. If the field was unified at the level of one theory, the unification would dissolve the very approaches it claims to connect. The unity of futures studies lies elsewhere. It lies in commitments that every ecosystem presupposes: that future possibilities are real and worth studying, that reasoning about them goes wrong in specific ways, and that people are always engaged with futures (Virmajoki, forthcoming).

The fifth role, theory about the field, is where the two questions of this essay begin to meet. Accounts of what futures studies is, and pictures such as the cone, are taught, repeated and used. They shape how practitioners see the space they study. Now, if the *possibility space* is reflexive, then the field's own accounts belong to that space. This is what I meant by saying that the lens is in the picture. In order to see what follows from this, we need to look more carefully at the idea that theories can shape what they describe.

## **6. When saying is doing**

The idea that describing something can change it is an old idea. It comes in several versions, and the versions are easily mixed with each other.

The oldest version is the notion of a *performative utterance* by J. L. Austin (1962). Some sentences do not describe a state of affairs. They bring it about: "I promise" makes a promise, and "I name this ship" names the ship. We should notice that there is no causal chain from belief to behaviour here. The saying constitutes the fact.

The second version is the *self-fulfilling prophecy* of Robert K. Merton (1948): a belief about a situation, false when it is formed, leads people to act in ways that make it come true. Merton's example is a bank run. There is a rumour that a sound bank is insolvent, depositors withdraw their money because of the rumour, and the bank fails. The prophecy was false when it was uttered, and it was true when everything was over.

The third version is what Karl Popper (1957, p. 13), who coined the term, called the *Oedipus effect*: a prediction can itself influence the event it predicts, by bringing it about or by preventing it. Popper discussed it in *The Poverty of Historicism*, his critique of the idea that history can be predicted like an eclipse; there it figures among the arguments offered for why social prediction cannot be exact. It has been a standing worry for social prediction since then. If the publishing of a forecast changes what people do, can a public forecast ever be correct? Economists gave a clever answer already in the 1950s. Grunberg and

Modigliani (1954) and Simon (1954) showed that, under reasonable conditions, there exists a prediction which is fulfilled once it has been published and reacted to – a *fixed point*, so to say. So reflexivity does not make correct public prediction impossible. What it changes is what a correct prediction is. A correct prediction is one that takes its own effects into account.

The fourth version interests me most, because it concerns theories and not individual predictions. Michel Callon (1998) proposed that economics does not simply describe an economy that exists already. Economics helps to bring the economy into being; it performs the economy. Donald MacKenzie developed the idea with unusual care, and he makes a series of distinctions. There is a theory that is merely used by participants (*generic performativity*). There is a theory whose use makes a difference (*effective performativity*). There is a theory whose use makes the world more like the picture the theory gives of it; this is *Barnesian performativity*, named after the account of self-validating social knowledge by Barry Barnes (1983). And finally there is *counterperformativity*, where the use makes the world less like the theory (MacKenzie 2006). As the title of MacKenzie's book puts it, a theory can be an engine and not only a camera.

A fifth relative comes from the philosophy of science itself. Ian Hacking (1999) distinguished “indifferent” kinds from “interactive” kinds. Quarks are of *indifferent kind*: they do not care how we classify them. Categories of people are interactive. The people classified may respond, directly or indirectly, to the classifications applied to them, and these responses can in turn change the classifications. Hacking (1995) called this the *looping effect*. Human kinds are moving targets.

These are not five names for one thing, and we need to keep them apart. Austin's performative constitutes. Merton's prophecy causes through belief. Popper's effect is any influence of a prediction on its outcome. In *Barnesian performativity*, the descriptive adequacy of a theory is produced by the use of the theory. And looping is about kinds that respond to being classified. Why do the differences matter? Because they imply different evidence and different evaluations.

The study of the Chicago Board Options Exchange by MacKenzie and Millo (2003) is one of the best cases I know where several of these mechanisms can be seen at work at once. When options trading began there in 1973, it had to overcome a moral problem. Options looked like gambling, and regulators treated them with suspicion. Economics helped to legitimate options by recasting them as instruments of efficient pricing. The *option pricing theory* of Black, Scholes and Merton was published in the same year, and at first it fitted market prices only roughly. Over the following years the fit became excellent.

However, the improvement did not mean that the theory had found regularities that were waiting to be discovered. The market itself was altered, and the alterations brought reality closer to the theory's assumptions: credit rules were eased for hedges that were defined with the theory, commissions were deregulated, and price information travelled faster. Traders began to use the theory, first on printed sheets and later in software, and its key parameter (implied volatility) became the language in which options were discussed and traded. The arbitrage of the traders, guided by the theory, removed exactly those deviations that the central econometric test measured. So the theory was, in part, helped to pass its own test. After the stock market crash of October 1987 the fit broke. The market began to price large falls as a real possibility, and that pattern persisted. It was sustained by memory, by margin rules, and by moral disapproval towards those who sold crash protection without covering it (see also MacKenzie 2006).

From this case we can read three ways in which a theory can change the world, and one way in which the world can push back. First, the theory can change how something is seen: legitimization. Second, the world can be adjusted to fit the assumptions of the theory: realisation. Third, people can use the theory so that reality moves closer to it: fit that is made by use. Fourth, an event can change what people treat as possible and undo the fit, as happened in the market after 1987, when crashes became thinkable and got a price. MacKenzie (2006, pp. 33–34) raises the possibility of *counterperformativity* here: portfolio insurance, which was built on option theory, may have contributed to the crash, though how much cannot be determined, and afterwards prices no longer followed the classic model. Even so, the theory stayed in the language of the market: implied volatility remained the terms in which the new fear of crashes was priced. Then again, MacKenzie and Millo are equally clear about the limits. The theory did not win by its elegance; the floor traders mocked the price sheets at first. It needed tools, it needed a network of trusted people who carried the authority of economics to regulators, and it needed time. It could be undone by an event. And the traders who made the theory true were not the cold calculators that the theory assumed. They relied on reputation, reciprocity and a shared sense of what was acceptable. The market was configured cognitively by the theory, but never morally in its image. Finally, there is a methodological lesson that I want to underline. The authors could see all this only because they followed the case over decades. A snapshot of the market at almost any single moment would have given the wrong answer to the question whether the theory had discovered the prices or produced them.

## **7. Is futures studies an engine?**

Futures studies has said for a long time that it is. Bell described the field as a practical discipline that aims to change the world and not only to describe it.

*Causal Layered Analysis* was designed to open transformative spaces for alternative futures. The whole thesis of Polak is that images of the future steer societies. Bell counts the *self-altering prophecy* among the key concepts of the field. The work on *futures literacy* speaks of learning to use the future in the present (Miller 2018), and the research on *techniques of futuring* examines how imagined futures become socially consequential (Oomen et al. 2022). So, in a sense, futures studies did not need economic sociology in order to notice performativity. Reflexivity is the field's own word for it: images of the future change present action, and present action changes which futures become possible. I have argued elsewhere that this omnipresence of futures in human life is one of the commitments that hold the field together (Virmajoki, forthcoming).

However, the self-description of a field is one thing and evidence is another thing. Here I want to be deliberately unromantic. Most things happen despite futures studies. Most scenario reports are read by a handful of people, and they change nothing that anyone could trace. When futures studies claims performativity for itself, it often does so without evidence. The effect is presumed, and because it is presumed, nobody looks for it. In a grim way, Devil's Island is a better model of real performativity than most foresight reports are. Its image of the future (nobody escapes) did not work because it was elegant. It did not even work because it was true. It worked because it was carried by something: by cliffs, by guards, by punishments for failed attempts, and by the expectations of the prisoners themselves. Images of the future perform when something carries them. So the honest question is not whether futures work can shape futures. Of course it can. The question is which of the theories, models and pictures of the field do shape futures, in which of the senses above, and how we could tell.

A first step is to notice that different kinds of claims about futures perform in different ways, and they also succeed in different ways. A forecast can fulfil itself or defeat itself, usually by accident, and its success condition is correspondence. A vision is meant to perform: it gathers allies and resources, and it succeeds if it pulls people towards it. A warning succeeds if the thing it warns about never happens. This means, odd as that may sound, that a successful warning fails as a prediction. A roadmap is a coordination device, as it lines up actions so that its steps come true. And a scenario does something that is more subtle still. It does not aim at an outcome at all. It acts on what people treat as possible, plausible and desirable.

With this last point we arrive at a kind of performativity that is distinctive to futures studies, and it is worth a name: *modal performativity*. A scenario set does not predict. What it does is to rearrange the modal landscape of its users, i.e., which futures they take seriously, which they dismiss, which they fear, and

which they begin to prepare for. The favourite example of the field is the scenario work of Shell in the early 1970s. Shell was ready for the 1973 oil crisis. In Wack's (1985) account, what mattered was less any forecast than a change in the managers' mental model: a future that did not match the prevailing expectation became one they took seriously. Notice what the example shows and what it does not show. It shows performativity inside one organisation, as a changed capacity to act. It does not show that scenarios changed the oil market. But the larger the effect that is claimed, the heavier the burden of evidence is.

Modal performativity cuts both ways. A scenario can open a space, and a dominant scenario can close one. When one image of the future dominates, it begins to look natural, and the alternatives drop out of view. This is the point of Polak turned around. The options market before 1987 is an uncomfortable illustration. The more the model was used, the better it fitted, and, arguably, the less anyone looked outside of it. A crash of the size that came was, on the model's own assumptions, almost beyond the conceivable (MacKenzie 2006). This is the dark side of successful performativity: an account that works can shrink the very space it describes. It is also one more reason why the *futures cone* misleads us. The space of the cone only widens. Real *possibility spaces* shrink too, and performed accounts are among the forces that shrink them.

Models perform in yet another way. A model tells what follows if its assumptions hold. However, when the assumptions are taken up as targets, the model can become a plan. Climate policy offers a more contested example. Critics have argued that *integrated assessment models* assumed large-scale removal of carbon dioxide from the atmosphere in the future, that this made ambitious temperature targets look more attainable than they are, and that the models thereby shaped what policy-makers treated as possible (Anderson & Peters 2016; Beck & Mahony 2018). I do not try to settle that debate here. The point is that it is a debate about performativity, and it has the right form. It asks which assumptions travelled from models into policy, through which channels, and with what effects.

And then there is the field's own theory. The *futures cone* is drawn in almost every foresight course. It suggests that the near future is settled and the far future is open, that options widen automatically and never shrink, and that one can read shares of plausibility at a given date. Now, if practitioners who are trained on the cone come to look for cone-shaped futures, then the cone performs a modal structure. This would be *modal performativity* that is produced by the field's own *domain theory*. I offer this as a hypothesis and not as a verdict. It could be studied, for example, by comparing how people who are trained in the cone and people who are not map the same situation. But the hypothesis makes visible what the reflexivity of the object implies. A field

whose object includes accounts of futures must count its own accounts among them. Again: the lens is in the picture.

## 8. Testing with coconuts

Papillon tested his conjecture before he trusted his life on it. He could do so because the sea is what Hacking would call an *indifferent kind*: throwing sacks into it does not change how the waves behave. Futures studies has that luxury rarely. When the object responds to being described, the description interferes with its own test. The difficulty has several parts, and it helps to discuss them one by one.

The first part is that fit is not discovery. If the use of a model can produce the fit between the model and the world, then the fit cannot, by itself, confirm the model. For roughly a decade, *option pricing theory* fitted observed option prices extraordinarily well, and part of that fit was produced by the users of the theory. The lesson is uncomfortable for anyone in futures studies who likes to cite success stories. A forecast that came true may have been right. It may have been lucky. Or it may have been self-fulfilling. The hit alone cannot tell which. Being lucky is not a quality criterion; not in futures studies, not anywhere.

The second part is the inverse one: success can be invisible. A warning that works prevents the thing it warns about. By the standard of predictive accuracy the warning failed, while by the standard of its purpose it succeeded. So any evaluation of futures work that counts hits and misses will systematically punish the most successful warnings. This is no mere technicality. It is one of the reasons why predictive accuracy cannot be the general quality criterion of futures studies. The quality of futures work has to be judged relative to what the work aims to do – to explore a space, to change judgement, to prepare, to prevent, and so on (Virmajoki, forthcoming).

The third part is that success breeds blindness. A performed account shapes the data that would test it, and it directs attention away from its alternatives. Kyle Stanford (2006) has argued that scientists have repeatedly failed to conceive alternatives to their best theories even though alternative theories, later accepted by scientific communities, could also have been well supported by the evidence available at the time. In futures studies the problem is sharper. Why? Because here the unconceived alternatives are futures, and a successful account can make them less likely as well as less visible.

How could we, then, ever know that a futures account performed? My answer is the answer a philosopher of causation would give. To say that a scenario, a model or a picture performed is to make a causal claim: had it been absent or different, things would have gone differently. On the *interventionist view of causation*, X makes a difference to Y if an intervention on X (changing it,



removing it) would change Y (Woodward 2003). Performativity claims have exactly this form, and therefore they need exactly this kind of support. A story of influence is not enough. We need evidence that bears on the counterfactual: who took up the account, where it was built into decisions, devices or rules, and what changed in what these people did. MacKenzie and Millo could answer such questions because they interviewed participants and followed prices, rules and technologies through time. Futures studies almost never does this for its own outputs. Rarely it follows a scenario set into the organisation that commissioned it and asks, years later, what difference the scenarios made. This is, I think, one of the most important empirical research programmes the field could adopt. It is also a place where the historical sciences have much to teach to the field, as they have methods for inferring the unobserved from traces (Currie 2018).

There is also a lesson about dissent. Institutions that must appear certain tend to bury disagreement. An image such as “nobody escapes” survives best where failed attempts are punished and doubts are not spoken aloud. Futures studies, at its best, does the opposite. Scenario work keeps several futures in view for the very reason that it resists the collapse into one future. Good Delphi practice keeps the spread of expert opinion and does not report only the consensus, because the pattern of disagreement may be the most informative thing the study produces. In both cases, *implicit problem recognition* is a recognition of what dissent is for. If futures accounts perform, then dissent is not noise that should be cleaned away before publication. It is the part of the account that keeps the space open.

## **9. Responsibility without grandiosity**

If accounts of the future can shape the future, then the producing of such accounts is a kind of action, and action is open to evaluation. Two consequences follow. Both are easily overstated, so I try to state them with care.

The first consequence is that values are not an external addition to futures research. They belong to its object. Desirable and alternative futures are layers of the *possibility space*, on the same footing with the probable and plausible ones. A *possibility space* without a desirability layer is therefore not a neutral description. It is an incomplete description. In any case, the ideal of value-free science has been strongly challenged in philosophy of science: Douglas (2009), for example, argues that values enter decisions about what evidence is sufficient and which errors are worth risking. Futures studies is simply more visibly value-laden than most fields, because its object explicitly includes what people take to be desirable. The methodological task is not to remove values.

The task is to make explicit which values do which work in which layer of the space.

The second consequence is a responsibility that comes with reflexivity. When a scenario becomes a reference point, or a vision attracts funding, or a threat image steers regulation, someone has chosen which futures to articulate, whose perspectives to include and whose to leave out. The leaving out of voices is then not only an ethical failure. It is an epistemic failure too. Consider an expert panel of people with similar backgrounds. Such a panel can produce a perfectly coherent foresight that systematically misses futures which are visible only from other positions: one sees efficiency where another sees dependence, one sees security where another sees surveillance. The concept that gives the argument precision is *epistemic injustice*, i.e., wronging someone as a knower (Fricker 2007). Fricker distinguishes two forms of this injustice. First, *testimonial injustice* occurs when a person is given less credibility than the person deserves, and the reason is a prejudice attached to the identity of that person. Second, *hermeneutical injustice* is subtler and harder to make visible. It occurs when the concepts available to a person are not sufficient for the person to make sense of, or to express conceptually, her own experience. Moreover, the insufficiency is not accidental: it reflects the unequal participation of social groups in the practices through which shared concepts are made. A participant who has been invited, and who is listened to attentively, may still find that what she needs to say about the future has no place to go in the templates that are offered. The vocabulary simply has no slot. In a reflexive field, whose accounts get made is part of whose futures get made.

However, responsibility must not turn into grandiosity. Futures studies does not control the future, and it should not create the impression that it does. This humility is not mere politeness. It is epistemically accurate. Most things happen despite the field, as I said above, and claims to the contrary are just claims until they are shown otherwise. The right attitude holds both thoughts at once. We should take seriously that our accounts may shape futures (usually modestly, occasionally decisively), and we should take equally seriously the task of finding out when they do. For practice, this can be put in one sentence: the question is not whether futures work affects the future, but whether its knowledge is justified and its values acceptable.

## **10. The caretakers**

In an earlier essay I borrowed another film, *The Shining* of Kubrick, to describe a risk for the philosophy of futures studies. The risk is to become trapped, like Jack Torrance in the Overlook Hotel, in endless writing that no longer touches the reality around it, i.e., in grand metaphysical questions that are pursued for their own sake (Virmajoki 2026a). I ended that essay by saying that we are the

caretakers and have always been. Philosophical reflection on futures studies is not something that is imported from outside of the field. It is something futures researchers have to do themselves.

This essay adds a twist to that. In Kubrick's film the line is sinister: the caretaker has always been inside the hotel, a part of its history, whether he knew it or not. This is the situation of futures studies too. Its researchers are not outside the *possibility space* looking in. Their scenarios, models, pictures and theories (including the theories of the field itself) are among the things that make up the space they study.

When we put the two halves of the essay together, we get a single view. The talk about theory in futures studies becomes less messy when we ask what role a theory plays: lens, input, theory of our object, theory built into methods, or theory of the field. The domain of the field is the *possibility space*, and the *possibility space* is reflexive. Therefore the theories of the field can perform. And when they do, the distinction between lens and object, which is so useful elsewhere, no longer holds cleanly. This is no scandal. It is the peculiar condition of a field whose object includes the field's own accounts. What the condition demands is not a grander theory. It demands a more careful practice: saying which kind of claim one is making, keeping dissent in view, and treating every claim of influence as a causal claim that needs evidence.

In the last shot of *Papillon*, the old convict drifts away from the island on his raft made of coconut sacks, and he shouts back at the island that he is still here. Futures studies has been told many times, and by serious people, that it has no object of its own and no theory worth the name. It is still here too. The task now is not to shout back. The task is to show, one careful study at a time, what the field studies and what its accounts do.

The future is a playground of the human mind, as a colleague and I once wrote, but it has to be handled one question at a time (Heinonen & Virmajoki 2025). This essay has asked two questions and found that they were one. There are many more questions, and to answer them properly is, to be honest, rather dull work. Still, it is the kind of work that leads somewhere.

## References

- Anderson, K., & Peters, G. (2016). The trouble with negative emissions. *Science*, 354(6309), 182–183.
- Austin, J. L. (1962). *How to do things with words*. Oxford: Clarendon Press.
- Barnes, B. (1983). Social life as bootstrapped induction. *Sociology*, 17(4), 524–545.
- Beck, S., & Mahony, M. (2018). The politics of anticipation: The IPCC and the negative emissions technologies experience. *Global Sustainability*, 1, e8.
- Bell, W. (2009 [1997]). *Foundations of futures studies: Human science for a new era. Vol. 1: History, purposes, and knowledge*. Transaction Publishers.
- Brier, D. J. (2005). Marking the future: A review of time horizons. *Futures*, 37(8), 833–848.
- Callon, M. (Ed.) (1998). *The laws of the markets*. Oxford: Blackwell.

- Currie, A. (2018). *Rock, bone, and ruin: An optimist's guide to the historical sciences*. MIT Press.
- Douglas, H. (2009). *Science, policy, and the value-free ideal*. University of Pittsburgh Press.
- Fernani, A., & Chermack, T. J. (2021). The resistance to scientific theory in futures and foresight, and what to do about it. *Futures & Foresight Science*, 3(3–4), e61.
- Fricker, M. (2007). *Epistemic injustice: Power and the ethics of knowing*. Oxford University Press.
- Grunberg, E., & Modigliani, F. (1954). The predictability of social events. *Journal of Political Economy*, 62(6), 465–478.
- Hacking, I. (1995). The *looping effects* of human kinds. In D. Sperber, D. Premack & A. J. Premack (Eds.), *Causal cognition: A multidisciplinary debate* (pp. 351–383). Oxford: Clarendon Press.
- Hacking, I. (1999). *The social construction of what?* Harvard University Press.
- Hancock, T., & Bezold, C. (1994). Possible futures, preferable futures. *Healthcare Forum Journal*, 37(2), 23–29.
- Heinonen, S., & Virmajoki, V. (2025). Future is philosophy! Core philosophical questions for futures studies. *Human Futures*, October 2025, 26–32.
- Inayatullah, S. (1998). *Causal Layered Analysis: Poststructuralism as method*. *Futures*, 30(8), 815–829.
- Kubrick, S. (Director). (1980). *The Shining* [Film]. Warner Bros.
- Le Bihan, S. (2017). Enlightening falsehoods: A modal view of scientific understanding. In S. R. Grimm, C. Baumberger, & S. Ammon (Eds.), *Explaining understanding: New perspectives from epistemology and philosophy of science* (pp. 111–136). Routledge.
- Lukka, K., & Vinnari, E. (2014). Domain theory and *method theory* in management accounting research. *Accounting, Auditing & Accountability Journal*, 27(8), 1308–1338.
- MacKenzie, D. (2006). *An engine, not a camera: How financial models shape markets*. MIT Press.
- MacKenzie, D., & Millo, Y. (2003). Constructing a market, performing theory: The historical sociology of a financial derivatives exchange. *American Journal of Sociology*, 109(1), 107–145.
- Merton, R. K. (1948). The *self-fulfilling prophecy*. *The Antioch Review*, 8(2), 193–210.
- Miller, R. (Ed.) (2018). *Transforming the future: Anticipation in the 21st century*. Routledge/UNESCO.
- Oomen, J., Hoffman, J., & Hajer, M. A. (2022). Techniques of futuring: On how imagined futures become socially performative. *European Journal of Social Theory*, 25(2), 252–270.
- Polak, F. (1973). *The image of the future* (E. Boulding, Trans. & abridged). Amsterdam: Elsevier.
- Popper, K. R. (1957). *The poverty of historicism*. London: Routledge & Kegan Paul.
- Schaffner, F. J. (Director). (1973). *Papillon* [Film]. Allied Artists.
- Simon, H. A. (1954). Bandwagon and underdog effects and the possibility of election predictions. *Public Opinion Quarterly*, 18(3), 245–253.
- Stanford, P. K. (2006). *Exceeding our grasp: Science, history, and the problem of unconceived alternatives*. Oxford University Press.
- Vaivio, J. (2008). Qualitative management accounting research: Rationale, pitfalls and potential. *Qualitative Research in Accounting & Management*, 5(1), 64–86.
- Virmajoki, V. (2026a). The shining: Illuminating philosophy and futures studies. *Futures*, 179, 103812.
- Virmajoki, V. (2026b). The greatest mystery of futures studies. *Futures & Foresight Science*, 8, e70046.
- Virmajoki, V. (forthcoming). On the intellectual unity of futures studies.
- Voros, J. (2003). A generic foresight process framework. *Foresight*, 5(3), 10–21.
- Wack, P. (1985). Scenarios: Uncharted waters ahead. *Harvard Business Review*, 63(5), 72–89.
- Woodward, J. (2003). *Making things happen: A theory of causal explanation*. Oxford University Press.
- Zimmerman, J. L. (2001). Conjectures regarding empirical managerial accounting research. *Journal of Accounting and Economics*, 32(1–3), 411–427.